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Problem Statement:

Use WordNet to identify semantic relationships such as synonymy, antonymy, and hypernymy in text data.

Objective:

- To understand the structure and organization of the WordNet lexical database.
- To extract synonyms of words by iterating over WordNet synsets and their lemmas.
- To identify antonyms of words using lemma antonym relationships in WordNet.
- To retrieve hypernyms of words to understand broader categorical relationships.
- To build a reusable function that returns all three semantic relations for any given English word.

S/W Packages and H/W apparatus used:

- Operating System: Windows/Linux/macOS
- Kernel: Python 3.x
- Tools: Jupyter Notebook / Anaconda
- Hardware: CPU with minimum 4GB RAM
- Libraries and packages used: NLTK, NLTK WordNet corpus (wordnet, omw-1.4)

Theory:

Lexical semantics is the study of word meanings and the relationships between them. Understanding these relationships is central to many NLP tasks such as word sense disambiguation, text similarity, and knowledge graph construction. WordNet is the most widely used lexical database for English that organizes words based on their meanings rather than their forms.

WordNet:

WordNet is a large lexical database of English developed at Princeton University. Words are grouped into sets of cognitive synonyms called synsets, each expressing a distinct concept. Synsets are interlinked by conceptual-semantic and lexical relations. NLTK provides direct access to WordNet via `nltk.corpus.wordnet`. The Open Multilingual Wordnet (omw-1.4) package extends this to multiple languages.

Synonymy:

Synonyms are words that share the same or very similar meaning. In WordNet, all lemmas within a synset are synonyms of each other for that particular sense. For example, the synset for the concept of a 'canine animal' includes 'dog', 'domestic_dog', and 'Canis_familiaris' as synonyms. A word may belong to multiple synsets (one per sense), so the full synonym set is the union of lemmas across all its synsets.

Antonymy:

Antonyms are words with opposite meanings. In WordNet, antonym relationships are stored at the lemma level rather than the synset level. Each lemma can have zero or more antonym lemmas. For example, the lemma 'good' in the adjective synset has the antonym 'bad'. Not all words have antonyms - concrete nouns like 'dog' or 'car' typically have none, while adjectives and some verbs commonly do.

Hypernymy:

Hypernymy represents an 'is-a' or generalization relationship between concepts. A hypernym is the broader, more general category to which a word belongs. For example, 'vehicle' is a hypernym of 'car', and 'animal' is a hypernym of 'dog'. Hypernyms are accessed at the synset level via `syn.hypernyms()`. The complete hierarchy of hypernyms from a word up to the root concept forms a taxonomy tree, which is central to WordNet's ontological structure.

Methodology:

- Install NLTK and download the required corpora: wordnet and omw-1.4.
- Import nltk and the wordnet corpus as `wn` from `nltk.corpus`.
- Define a `get_semantic_relations(word)` function that initializes empty sets for synonyms, antonyms, and hypernyms.
- Iterate over all synsets of the input word using `wn.synsets(word)`.
- For each synset, iterate over its lemmas: add each lemma name to the synonyms set, and if a lemma has antonyms, add the first antonym's name to the antonyms set.
- For each synset, iterate over `syn.hypernyms()` and extract the base word (before the first dot) from the synset name to add to the hypernyms set.
- Return a dictionary containing the word and its three relation lists.
- Test the function on three sample words - 'good', 'dog', and 'car' - printing the top 5 results for each relation.

Use Cases:

- Use Case 1 - Query Expansion in Search Engines: Synonyms retrieved from WordNet are used to expand search queries, allowing a search for 'automobile' to also return results containing 'car' or 'vehicle', improving recall in information retrieval systems.
- Use Case 2 - Sentiment Analysis Enhancement: Antonym relationships help sentiment classifiers understand polarity shifts. Knowing that 'good' and 'bad' are antonyms allows models to correctly interpret negation patterns like 'not good' as negative sentiment.
- Use Case 3 - Text Similarity and Semantic Reasoning: Hypernymy chains enable semantic similarity computation between words. Two words that share a close common hypernym (e.g., 'dog' and 'cat' both under 'animal') are treated as more similar than words with distant hypernyms, useful in question answering and textual entailment systems.

Advantages:

- WordNet provides a rich, manually curated semantic network covering over 155,000 English words across 175,000 synsets.
- No training data or machine learning required; semantic relations are directly accessible via NLTK's API.
- Supports multiple levels of semantic granularity -from fine-grained synonyms to broad hypernym taxonomies.
- The omw-1.4 extension enables multilingual semantic relation queries beyond English.

- The `get_semantic_relations()` function is fully reusable and extensible to include additional relations like hyponyms, meronyms, and holonyms.

Limitations:

- WordNet does not capture contextual or domain-specific word senses; 'bank' returns relations for all senses (financial institution and riverbank) without disambiguation.
- Antonym coverage is limited primarily to adjectives; many nouns and verbs have no antonyms recorded in WordNet.
- WordNet is manually constructed and may lack coverage for technical jargon, neologisms, proper nouns, and informal language.
- Synonym sets from all senses are merged without disambiguation, which can introduce noise (e.g., 'dog' as a verb meaning 'to follow' appears alongside 'dog' as an animal).
- WordNet's hierarchical structure is based on formal English; it may not reflect everyday language usage or cultural variations.

Applications:

- Natural language understanding: WordNet semantic relations enrich chatbot and virtual assistant knowledge bases for better intent recognition.
- Automated thesaurus generation: Synonym and antonym extraction powers writing assistance tools and paraphrase generation systems.
- Knowledge graph construction: Hypernymy chains provide taxonomic backbone for ontologies used in semantic web and linked data applications.
- Word sense disambiguation (WSD): Hypernym and synonym relations provide contextual clues to identify the correct sense of ambiguous words in text.
- Educational NLP tools: Vocabulary learning applications use WordNet to generate synonym quizzes, antonym exercises, and word classification tasks.

Conclusion:

This assignment demonstrated the use of NLTK's WordNet interface to extract three fundamental semantic relationships -synonymy, antonymy, and hypernymy -for any given English word. By iterating over synsets and lemmas, we retrieved synonym sets representing shared meanings, antonyms representing opposite meanings, and hypernyms representing broader categorical concepts. The results for 'good', 'dog', and 'car' illustrated how these relations vary by word type: adjectives like 'good' yield rich antonym sets, while concrete nouns like 'dog' and 'car' provide extensive hypernym hierarchies. WordNet remains an invaluable resource for building semantically aware NLP systems without requiring large training corpora.